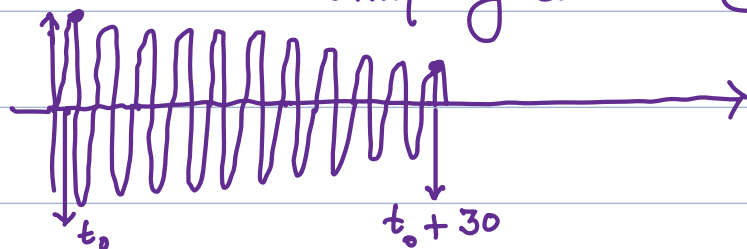


MTH 204 : Lecture 08

Ex: Consider an underdamped motion of a body of mass $m = 2\text{ kg}$. If the time between two consecutive maxima is 2 sec. and the maximum amplitude decreases to $\frac{1}{4}$ of its initial value after 15 cycles, what is the damping constant (c) of the system?



Here $m = 2\text{ kg}$, $T = 2\text{ sec}$

Let maximum amplitude occurs at t_0 .

Then $\frac{1}{4}$ (max amplitude) occurs at $t_0 + 30$

(Since $T = 2\text{ sec}$, time taken for 15 cycles is 30 sec)

Now the equation in this case is $y(t) = \tau e^{-\alpha t} \cos(\omega^* t - \delta)$

Now maximum amplitude $= \tau e^{-\alpha t_0}$

$$\frac{1}{4} (\text{max amplitude}) = \frac{1}{4} \tau e^{-\alpha t_0} = \tau e^{-\alpha (t_0 + 30)}$$

$$\Rightarrow \frac{1}{4} \tau e^{-\alpha t_0} = \tau e^{-\alpha t_0} e^{-30\alpha} \Rightarrow e^{-30\alpha} = \frac{1}{4}$$

$$\Rightarrow 30\alpha = \ln 4 \Rightarrow \alpha = \frac{\ln 4}{30} = \frac{2 \ln 2}{30} = \frac{\ln 2}{15}$$

If c is the damping constant, $\alpha = \frac{c}{2m} \Rightarrow \frac{\ln 2}{15} = \frac{c}{2m}$

$$\Rightarrow c = 2 \cdot m \frac{\ln 2}{15} = 2 \times 2 \frac{\ln 2}{15} = \boxed{\frac{4 \ln 2}{15}}$$

$$\Rightarrow c = .1848$$

Non homogeneous ODEs (NH):

$$y'' + p(x)y' + q(x)y = r(x) \text{ is (NH)}$$

A general solution of the NH ODE is of the form $y_h + y_p$ where y_h is the general solution of the homogeneous equation

$$y'' + p(x)y' + q(x)y = 0 \quad (H)$$

and y_p is a particular solution of (NH)

Theorem 1: If y_H is a solution of H and y_N is a solution of NH, then $y_H + y_N$ is a solution of NH.

Proof:

$$\begin{aligned} & (y_H + y_N)'' + p(x)(y_H + y_N)' + q(x)(y_H + y_N) \\ &= (y_H'' + p(x)y_H' + q(x)y_H) + (y_N'' + p(x)y_N' + q(x)y_N) \\ &= r(x) + 0 = r(x) \end{aligned}$$

Theorem 2: If y_1 and y_2 are two solutions of NH, then $y_1 - y_2$ is a solution of H.

Proof:

$$(y_1 - y_2)'' + p(x)(y_1 - y_2)' + q(x)(y_1 - y_2)$$

$$= \left[y_1'' + p(x)y_1' + q(x)y_1 \right] - \left[y_2'' + p(x)y_2' + q(x)y_2 \right]$$

$$= r(x) - r(x) = 0$$

Method of Undetermined coefficients:

Ex: $y'' - 3y' + 2.25y = 27(x^2 - x)$; $y(0) = 20$, $y'(0) = 30$

H: $y'' - 3y' + 2.25y = 0$

characteristic Equation: $\lambda^2 - 3\lambda + 2.25 = 0 \Rightarrow (\lambda - 1.5)^2 = 0$

So, there is a real double root $m = 1.5$

So, $y_h = (c_1 + c_2 x)e^{1.5x}$

Let $y_p = Ax^2 + Bx + C$ (where A, B, C are constants)

$$y_p' = 2Ax + B$$

$$y_p'' = 2A$$

So, if y_p is a solution of the NH,

then $2A - 3(2Ax + B) + 2.25(Ax^2 + Bx + C) = 27x^2 - 27x$

$$\Rightarrow 2.25Ax^2 + (2.25B - 6A)x + (2A - 3B + 2.25C) = 27x^2 - 27x$$

$$\Rightarrow 2.25A = 27 \Rightarrow A = \frac{27}{2.25} = \frac{27}{9} \times 4 = \boxed{12}$$

$$2.25B - 6A = -27 \Rightarrow 2.25B - 6 \times 12 = -27 \Rightarrow B = \frac{45}{2.25} = \frac{45}{9} \times 4 = \boxed{20}$$

$$2A - 3B + 2.25C = 0 \Rightarrow 2 \times 12 - 3 \times 20 + 2.25C = 0 \Rightarrow C = \frac{36}{2.25} = \frac{36}{9} \times 4 = \boxed{16}$$

Hence $y_p = 12x^2 + 20x + 16$

check: $y_p' = 24x + 20$, $y_p'' = 24$

Now $y_p'' - 3y_p' + 2.25y_p = 24 - 3(24x + 20) + 2.25(12x^2 + 20x + 16)$

~~$= 24 - 72x - 60 + 27x^2 + 45x + 36$~~

$= 27x^2 - 27x = \text{R.H.S of the NH}$

So, the general solution of the NH

is: $y(x) = y_h + y_p = (c_1 + c_2 x)e^{1.5x} + 12x^2 + 20x + 16$

Now $y(0) = 20 \Rightarrow c_1 e^0 + 16 = 20 \Rightarrow c_1 = 4$

$y'(x) = 1.5c_1 e^{.5x} + c_2 e^{1.5x} + 1.5c_2 x e^{.5x} + 24x + 20$

Now $y'(0) = 30 \Rightarrow 1.5(4)e^0 + c_2 e^0 + 20 = 30$

$\Rightarrow 6 + c_2 + 20 = 30$

$\Rightarrow c_2 = 4$

So, the particular solution of the Non homogeneous equation is:

$y(x) = [4 + 4x]e^{1.5x} + 12x^2 + 20x + 16$

or $y(x) = 4[(1+x)e^{1.5x} + 3x^2 + 5x + 4]$

Ex: $y'' - 2y' = 12e^{2x} - 8e^{-2x}$, $y(0) = -2$, $y'(0) = 12$

H: $y'' - 2y' = 0$

Characteristic equation: $\lambda^2 - 2\lambda = 0 \Rightarrow \lambda(\lambda - 2) = 0$

$\Rightarrow \lambda = 2, 0$

So, $y_h = c_1 e^{2x} + c_2 e^{0 \cdot x} = c_1 e^{2x} + c_2$

Let $y_p = Ax e^{2x} + B e^{-2x}$

Then $y_p' = A e^{2x} + 2Ax e^{2x} - 2B e^{-2x}$

and $y_p'' = 2A e^{2x} + 2A e^{2x} + 4Ax e^{2x} + 4B e^{-2x}$
 $= 4A e^{2x} + 4Ax e^{2x} + 4B e^{-2x}$

Note that e^{2x} is a solution of the homogeneous.

equation corresponding to a simple root since $4e^{2x} - 2(2e^{2x}) = 0$

Now if y_p is a solution of the NH, then

$$4A e^{2x} + 4Ax e^{2x} + 4B e^{-2x} - 2(A e^{2x} + 2Ax e^{2x} - 2B e^{-2x}) = 12e^{2x} - 8e^{-2x}$$

$$\Rightarrow 4A e^{2x} + \cancel{4Ax e^{2x}} + 4B e^{-2x} - 2A e^{2x} - \cancel{4Ax e^{2x}} + 4B e^{-2x} = 12e^{2x} - 8e^{-2x}$$

$$\Rightarrow 2A e^{2x} + 8B e^{-2x} = 12e^{2x} - 8e^{-2x}$$

$$\Rightarrow 2A = 12 \text{ and } 8B = -8$$

$$\Rightarrow A = 6 \text{ and } B = -1$$

Hence $y_p = 6x e^{2x} - e^{-2x}$

So, the general solution of the Nonhomogeneous Equation

is: $y(x) = c_1 e^{2x} + c_2 + 6xe^{2x} - e^{-2x}$

$$y'(x) = 2c_1 e^{2x} + 6e^{2x} + 12xe^{2x} + 2e^{-2x}$$

Given that $y(0) = -2$ and $y'(0) = 12$

Then $c_1 + c_2 - 1 = -2 \Rightarrow c_1 + c_2 = -1 \dots \textcircled{1}$

and $2c_1 + 6 + 2 = 12 \Rightarrow 2c_1 = 12 - 8 \Rightarrow 2c_1 = 4$
 $\Rightarrow \boxed{c_1 = 2}$

Substituting in $\textcircled{1}$ we get $2 + c_2 = -1$
 $\Rightarrow \boxed{c_2 = -3}$

Hence the particular solution of the Nonhomogeneous equation is:

$$\boxed{y = y(x) = 2e^{2x} - 3 + 6xe^{2x} - e^{-2x}}$$

Method of Undetermined Coefficient:

To solve $y'' + ay' + by = r(x)$

Term in $r(x)$

$$k e^{rx}$$

$$k x^n \quad (n=0,1,2,\dots)$$

$$\left. \begin{array}{l} k \cos \omega x \\ k \sin \omega x \end{array} \right\}$$

$$\left. \begin{array}{l} k e^{\alpha x} \cos \omega x \\ k e^{\alpha x} \sin \omega x \end{array} \right\}$$

Choice for $y_p(x)$

$$C e^{rx}$$

$$C_n x^n + C_{n-1} x^{n-1} + \dots + C_1 x + C_0$$

$$C_1 \cos \omega x + C_2 \sin \omega x$$

$$e^{\alpha x} (C_1 \cos \omega x + C_2 \sin \omega x)$$

Modification: If the term in $r(x)$ is

also a solution of H , then multiply by x or x^2 depending on if this solution corresponds to a simple root or a double root of the characteristic polynomial.

Sum Rule

If $r(x)$ is a sum of functions, choose a sum of y_p 's.

Example:

(1) If $r(x) = 10 \sin 2x$, then $y_p = A \sin 2x + B \cos 2x$

(2) $r(x) = 8 \cos x$
and it is a solution of H and it corresponds to a single root of the characteristic equation } then $y_p = x (A \cos x + B \sin x)$

(3) $r(x) = e^{2x} \sin 5x$
and it is a solution of H and it corresponds to a double root of the characteristic equation } then $y_p = x^2 e^{2x} (A \cos 5x + B \sin 5x)$

(4) $r(x) = \cos x + x^2 - 1$, then $y_p = A \cos x + B \sin x + Cx^2 + Dx + E$

(5) $r(x) = \cos x + x^2 - 1$ } then $y_p = x^2 (A \cos x + B \sin x)$
and $\cos x$ is a solution of H and it corresponds to a double root of the characteristic equation } $+ (Cx^2 + Dx + E)$

Ex: $y'' + 2y' + .75y = 2 \cos x - .25 \sin x + .09x$
 $y(0) = 2.78$, $y'(0) = -0.43$

For the general solution of the Homogeneous ODE,
we get the characteristic equation as:

Ex: $y'' + 2y' + .75y = 2\cos x - .25\sin x + .09x$, $y(0) = 2.78$, $y'(0) = -.43$

H: $y'' + 2y' + .75y = 0$

characteristic Equation: $\lambda^2 + 2\lambda + .75 = 0 \Rightarrow \lambda^2 + 2\lambda + \frac{3}{4} = 0$

$\Rightarrow (\lambda + \frac{1}{2})(\lambda + \frac{3}{2}) = 0 \Rightarrow \lambda = -\frac{1}{2}, -\frac{3}{2}$

So, $y_h = c_1 e^{-\frac{1}{2}x} + c_2 e^{-\frac{3}{2}x}$

Let $y_p = C \cos x + D \sin x + Ax + B$

Then $y_p' = -C \sin x + D \cos x + A$

$y_p'' = -C \cos x - D \sin x$

Since y_p is a solution of the N.H,

$-C \cos x - D \sin x - 2C \sin x + 2D \cos x + 2A + .75C \cos x$

$+ .75D \sin x + .75Ax + .75B$

$= 2\cos x - .25\sin x + .09x$

$\Rightarrow (-.25C + 2D) \cos x + (-2C - .25D) \sin x$

$+ .75Ax + (2A + .75B) = 2\cos x - .25\sin x + .09x$

$\Rightarrow -.25C + 2D = 2 \Rightarrow -.5C + 4D = 4$

$-2C - .25D = -.25 \Rightarrow -.5C - .625D = -.625$

$\begin{array}{r} -.5C + 4D = 4 \\ -.5C - .625D = -.625 \\ \hline 4.625D = 4.625 \Rightarrow \boxed{D=1} \end{array}$

Then $\boxed{C=0}$, Also $.75A = .09$ and $2A + .75B = 0$

$\Rightarrow A = \frac{.09}{.75} = \boxed{.12}$ and $B = -\frac{2 \times .12}{.75} = -\frac{.24}{.75} = \boxed{-.32}$

Hence the general solution of the ODE is:

$y(x) = c_1 e^{-\frac{1}{2}x} + c_2 e^{-\frac{3}{2}x} + \sin x + .12x - .32$

Now $y'(x) = -\frac{c_1}{2} e^{-\frac{1}{2}x} - \frac{3c_2}{2} e^{-\frac{3}{2}x} + \cos x + .12$

$$y(0) = 2.78 \text{ and } y'(0) = .43 \Rightarrow c_1 + c_2 - .32 = 2.78$$

$$\text{and } -\frac{c_1}{2} - \frac{3c_2}{2} + 1 + .12 = -.43$$

$$\Rightarrow c_1 + c_2 = 3.10$$

$$\Rightarrow -c_1 - 3c_2 = -3.10$$

$$\underline{-2c_2 = 0} \Rightarrow c_2 = 0 \text{ and hence } c_1 = 3.10$$

Hence the particular solution of the Nonhomogeneous Equation is:

$$y(x) = 3.10 e^{-\frac{1}{2}x} + \sin x + .12x - .32$$

Multiplication Rule:

If $r(x)$ is a product of functions in the table then the choice for y_p is also product of individual choices for each term in the product.

Ex: $r(x) = x e^{4x}$

Case 1: e^{4x} is not a solution of H : $y_p(x) = (Ax+B)e^{4x} = (Acx+Bc)e^{4x} = c_1 x e^{4x} + c_2 e^{4x}$

Case 2: e^{4x} is a solution of H : $y_p(x) = Ax^2 e^{4x} + Bx e^{4x}$

Case 3: e^{4x} and $x e^{4x}$ are both solutions of H : $y_p(x) = Ax^3 e^{4x} + Bx^2 e^{4x}$

Ex: $r(x) = (9x^2 - 103x) \cos x$

Case 1: $\cos x$ is not a solution of H : $y_p(x) = (Ax^2 + Bx + C)(D \cos x + E \sin x)$

$$= ADx^2 \cos x + AE x^2 \sin x + BDx \cos x + BE x \sin x + CD \cos x + CE \sin x$$

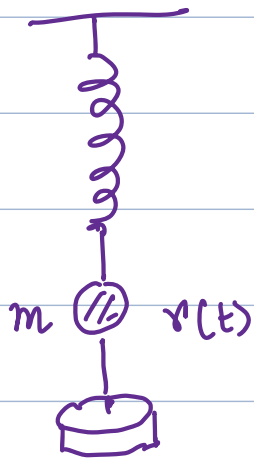
$$= A_1 x^2 \cos x + B_1 x^2 \sin x + C_1 x \cos x + D_1 x \sin x + E_1 \cos x + F_1 \sin x$$

Case 2: $\cos x$ is a solution of H :

$$y_p(x) = A_1 x^3 \cos x + B_1 x^3 \sin x + C_1 x^2 \cos x + D_1 x^2 \sin x + E_1 x \cos x + F_1 x \sin x$$

Forced Oscillation:

So far, we have considered vertical motion of a mass-spring system in the case of free motion:



That is no external force

but only internal forces controlled the motion.

These forces are my'' ,
damping force cy' ($c > 0$)
and Spring force ky (restoring force)

The ODE for this model is the homogeneous equation: $my'' + cy' + ky = 0$

Now we include an external force $r(t)$ and the resulting motion is called a Forced motion. Then the ODE will be $my'' + cy' + ky = r(t)$

In this case $r(t)$ is known as input or driving force

and $y(t)$ is called the output or response of the system to the driving force.

Let us assume $x(t) = F_0 \cos \omega t$ (Periodic
driving
force)
($F_0 > 0, \omega > 0$)

Then the ODE is:

$$m\ddot{y} + c\dot{y} + ky = F_0 \cos(\omega t)$$

Consider the associated

Homogeneous system: The roots of the characteristic

equation are $\lambda_1, \lambda_2 = -\frac{c}{2m} \pm \frac{1}{2m} \sqrt{c^2 - 4mk}$

Let y_h be the general solution of the homogeneous equation
We have seen different forms of y_h depending on the sign
of $c^2 - 4mk$.

Now for the particular solution of the
Non-homogeneous equation,

let $y_p = a \cos \omega t + b \sin \omega t$

Then $y_p' = -a\omega \sin \omega t + b\omega \cos \omega t$

and $y_p'' = -a\omega^2 \cos \omega t - b\omega^2 \sin \omega t$

Substituting :

$$m(-a\omega^2 \cos \omega t - b\omega^2 \sin \omega t) + c(-a\omega \sin \omega t + b\omega \cos \omega t) + k(a \cos \omega t + b \sin \omega t) = F_0 \cos \omega t$$

$$\Rightarrow (-m a \omega^2 + b c \omega + k a) \cos \omega t + (-m b \omega^2 - c a \omega + k b) \sin \omega t = F_0 \cos \omega t$$

$$\Rightarrow \left. \begin{aligned} -m a \omega^2 + b c \omega + k a &= F_0 \\ -m b \omega^2 - c a \omega + k b &= 0 \end{aligned} \right\}$$

$$\Rightarrow \left. \begin{aligned} (k - m \omega^2) a + (c \omega) b &= F_0 \\ (-c \omega) a + (k - m \omega^2) b &= 0 \end{aligned} \right\}$$

$$\Rightarrow \begin{pmatrix} k - m \omega^2 & c \omega \\ -c \omega & k - m \omega^2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} F_0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \left. \begin{aligned} a &= F_0 \frac{k - m \omega^2}{(k - m \omega^2)^2 + \omega^2 c^2} \\ b &= F_0 \frac{c \omega}{(k - m \omega^2)^2 + \omega^2 c^2} \end{aligned} \right\}$$

Recall $\omega_0^2 = \frac{k}{m} \Rightarrow k = m\omega_0^2$

$$\left. \begin{aligned} \text{So, } a &= F_0 \frac{m(\omega_0^2 - \omega^2)}{m^2(\omega_0^2 - \omega^2)^2 + \omega^2 c^2} \\ b &= F_0 \frac{c\omega}{m^2(\omega_0^2 - \omega^2)^2 + \omega^2 c^2} \end{aligned} \right\}$$

$$\begin{aligned} \text{So, } y_p &= F_0 \frac{m(\omega_0^2 - \omega^2)}{m^2(\omega_0^2 - \omega^2)^2 + \omega^2 c^2} \cos(\omega t) \\ &+ F_0 \frac{c\omega}{m^2(\omega_0^2 - \omega^2)^2 + \omega^2 c^2} \sin(\omega t) \end{aligned}$$

The general solution of the non-homogeneous equation

$$y(t) = y_h + y_p$$

Case 1: Undamped Forced Oscillation : ($c = 0$)

When $c = 0$

$$y_p = \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos(\omega t)$$

provided $\omega \neq \omega_0 = \sqrt{\frac{k}{m}}$
i.e. $\cos \omega t$ is not a solution
of the homogeneous equation
 $my'' + ky = 0$

Also the general solution of the homogeneous equation: $y_h = C \cos(\omega_0 t - \delta)$

The general solution of the non-homogeneous equation

is: $y = c \cos(\omega_0 t - \delta) + \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos(\omega t)$

(This is a superposition of two harmonic oscillations)

(This is valid as long as $\omega \neq \omega_0$)

Let us choose c and δ in such a way that

$$c = \frac{F_0}{m(\omega_0^2 - \omega^2)} \text{ and } \delta = 0$$

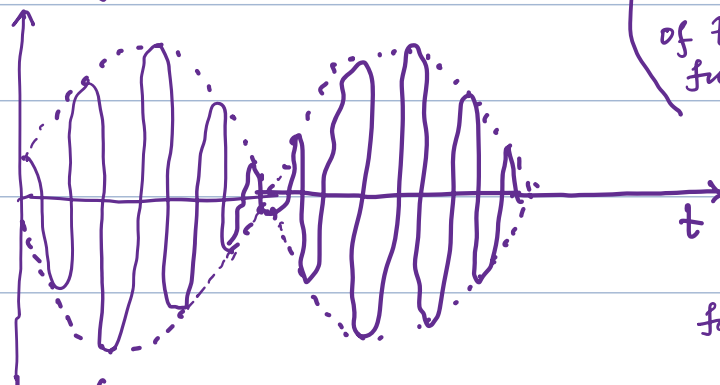
Then

$$y = \frac{F_0}{m(\omega_0^2 - \omega^2)} [\cos(\omega_0 t) + \cos(\omega t)]$$

$$= \left[\frac{2 F_0}{m(\omega_0^2 - \omega^2)} \cos\left(\frac{\omega_0 + \omega}{2} t\right) \cos\left(\frac{\omega_0 - \omega}{2} t\right) \right]$$

If $\omega_0 \approx \omega$, then we get a solution like:

(If $\omega_0 - \omega$ is small, the period of the last sine function is large)



The dashed curve results from the first cosine factor

(These solutions are called "beats" and this is what a musician hears when they tune their instruments)

(Forced undamped oscillation when the difference of the input and natural frequencies is small ("beats"))